

Class in JavaScript

Assignment Questions



TASK 1

Problem Statement 1: Create a Constructor for a Person

Write a JavaScript function constructor named **Person** that takes two parameters, name, and age, and Add a method to the prototype to display **name and age**. Then, create two instances of Person and display their names and ages.

TASK 2

Problem Statement 2: Implement a Bank Account

Create a constructor function named **BankAccount** that initializes a bank account with an initial **balance**. Include methods for depositing and withdrawing money from the account. Write code to demonstrate these operations on a bank account object.

TASK 3

Problem Statement 3: Create a Constructor for a Book

Design a constructor function called **Book** that takes **title and author** as parameters. Add a method to the prototype of the Book that displays the book's information. Create at least two book instances and display their information.

TASK 4

Problem Statement 4: Implement Task 1 using Class.

Design a JavaScript class called **Person** with properties for **name and age**. Implement a method to display the person's name and age. Then, create instances of Person and display their information.

TASK 5

Problem Statement 5: Implement a Calculator Class

Create a class called Calculator that initializes two values value1 and value2 to store numbers. Add methods for add, subtract, multiply, and divide. Perform and show operations.

TASK 6

Problem Statement 6: Design a Class for a Geometric Shape

Design a class called **Shape** that can represent various geometric shapes. Implement subclasses for specific shapes like **Circle, Rectangle, and RightTriangle**. Each subclass should have properties of that shape (e.g., Width, height for a rectangle) and methods for calculating the area and perimeter of the shape. Create instances of these shapes and calculate their areas and perimeters.

TASK 7

Problem Statement 7: Implement a Library System

Create a class called **Library** that can manage a collection of books. Each book should be represented as an instance of a **Book** class (with properties like title, author, and availability). Implement methods in the Library class to add books, check out books, and return books. Keep track of the books available.